

The Role of Depression in Relationships between Reappraisal Affordances, Choice, and Success

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Objective: The choice of emotion regulation (ER) strategies depends on contextual characteristics, and the flexible choice of ER strategies based on context is essential for psychological health, including depression. Focusing on reappraisal, this study examined how perceived reappraisal affordances (the inherent potential of a situation for semantic reappraisal)—a reappraisal-related situational factor—are linked to the success of reappraisal use via reappraisal choice and investigated the role of depression in this mediation.

Methods: Eighty participants (40 individuals with major depressive disorder [MDD] and 40 non-depressed controls) read eight vignettes describing negative-emotion-evoking situations and reported reappraisal affordance, the likelihood of choosing reappraisal, and negative emotional responses for each vignette. Then, they were asked to implement reappraisal for each situation, followed by reporting their negative emotional responses again.

Results: Reappraisal affordance predicted greater reductions in negative emotions following reappraisal use (i.e., greater success) via an increased choice of the strategy. Group moderated the affordance-choice relationship, with a significant positive relationship observed only in the MDD group, while the relationship was not significant in the controls.

Conclusion: These results support the idea that considering contextual factors is crucial in understanding ER; however, adjusting ER choice based on situational factors may not always reflect flexible ER, as indicated by the significant affordance-choice link only in the MDD group.

KEY WORDS: Reappraisal affordances; Emotion regulation; Depressive disorder, major; Emotions.

INTRODUCTION

Cognitive reappraisal is a widely employed emotion regulation (ER) strategy in which a person attempts to view an emotion-evoking situation more flexibly by altering their initial appraisal in order to reduce negative emotions [1]. According to the process model of ER [1,2], antecedent-focused strategies, such as reappraisal, which aim to modify emotional responses before they fully emerge, are considered more effective at reducing negative emotions than strategies used after emotions have fully developed. Indeed, prior studies have consistently demonstrated that habitual use of reappraisal is generally

associated with lower levels of various psychological difficulties, including depression [3]. However, recent research revealed that the choice and effectiveness of reappraisal depend on situational characteristics. For instance, previous studies using non-clinical samples demonstrated that reappraisal was more likely chosen in situations perceived less emotionally intensive [4] and using reappraisal in situations perceived as uncontrollable was associated with greater emotional well-being, whereas in situations perceived as controllable, reappraisal did not show the same beneficial effects [5].

Despite prior evidence for the context-dependent adaptability of reappraisal, research exploring situational/environmental factors associated with reappraisal remains limited. Among the limited factors studied, Suri *et al.* [6] proposed a situational factor specifically related to reappraisal, reappraisal affordances. Finding alternative interpretations is essential for reappraisal, but the feasibility or difficulty of finding alternative interpretations of a sit-

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uation can vary depending on the context of the situation. To reflect these differences across situations, Suri *et al.* [6] coined the term “reappraisal affordances,” which refers to the inherent potential of a situation and/or stimulus for semantic reappraisal. Suri *et al.* [6] argued that situations may offer different opportunities to re-interpret their possible meaning—that is different levels of reappraisal affordances—and such differences cannot be fully accounted for by the intensity of a situation (e.g., how negative a situation is). For example, an image depicting a person enduring a distressing medical procedure in a hospital may allow for several opportunities for reappraisal (e.g., the person will soon recover), whereas an image of deceased or mutilated children may provide few such opportunities. In this case, the former image has greater reappraisal affordances than the latter.

Given that reappraisal, particularly its flexible use depending on the characteristics of a given situation, is generally considered adaptive for emotional well-being [1, 7,8], it is important to investigate whether reappraisal affordances, a reappraisal-related situational factor, are linked to the effectiveness of reappraisal. This study aimed to examine whether reappraisal affordances predict reappraisal implementation success (i.e., decreases in negative emotions) via the increased choice of reappraisal. Furthermore, researchers have suggested that individual differences exist in reappraisal affordances and their link to choice [6,9], and that flexible ER is key to psychological health [7,8]. Thus, we sought to elucidate the role of depression (i.e., whether a person has a major depressive disorder [MDD]) in the model linking reappraisal affordances to reappraisal success.

Reappraisal Affordances, Choice, and Success

According to Tamir [10], ER involves effortful and motivated processes in which the decision to choose an ER strategy are influenced by an individual’s analysis of its costs and benefits, including the perceived effort required to implement the strategy. A person’s perception of the reappraisal affordances of a situation—how easy it is to use the strategy in a given context—could provide information about the cost or perceived effort required to use the strategy in that context, thereby influencing their intention and motivation to choose and implement it. Indeed, prior research has demonstrated that higher perceived reappraisal affordances predicted the greater choice of reappraisal,

even after accounting for the emotional intensity of stimuli [6,9,11].

In addition, self-determination theory [12] posits that one factor increasing individuals’ motivation to carry out a task is their perceived competence in successfully performing it, and this increased internal willingness, in turn, enhances performance outcomes. Previous studies have shown that both internal and external factors can influence a person’s perceived competence [13]. Based on self-determination theory, reappraisal affordances can serve as a situational factor closely related to one’s perceived competence in reappraisal, which in turn influences the choice and willingness to use reappraisal, ultimately affecting its success. That is, individuals are more likely to choose reappraisal in situations where they believe they can more easily employ reappraisal (higher reappraisal affordances). This increased willingness to choose reappraisal will in turn increase the likelihood of success, as assessed by a decrease in negative emotions, due to heightened motivation. Hence, we hypothesized that reappraisal affordances would predict reappraisal success via increased reappraisal choice.

The Role of Depression

We further investigated the role of depression in the proposed mediation model, predicting that the relationships among reappraisal affordances, choice, and success would be attenuated in individuals with MDD compared to controls. Theories highlighting ER flexibility argue that the ability to adaptively use ER strategies, based on varying situational characteristics and demands, is more important than the absolute frequency of using “putatively helpful” ER strategies for psychological health [7,8]. Bonanno and Burton [8] proposed context sensitivity, along with ER repertoire and response to feedback, as one of the three key components of ER flexibility. Context sensitivity refers to an individual’s ability to flexibly choose and employ a regulatory strategy based on the changing characteristics of situations. Researchers have argued that emotion dysregulation, such as depression, may arise from context-insensitive use of ER, leading to reduced adaptability to situations [7,8].

Accumulated research on ER in depression appears to support the view of ER flexibility [14-16]. Despite a consistent negative correlation between habitual reappraisal use, as measured by self-report questionnaires, and de-

pressive symptoms across studies [3], previous laboratory studies have found no significant differences in the choice/use and success of cognitive reappraisal between individuals with and without depression [15,16]. Instead, recent research suggests that depression could be more closely associated with difficulties in flexibly using ER strategies in response to varying situational demands [5, 14,17,18]. For instance, Chen and Bonanno [14] showed that individuals with more flexible ER patterns (e.g., greater sensitivity to contextual cues in the choice and use of ER strategies) tend to report lower levels of depressive symptoms. In addition, Haines *et al.* [5] found that individuals with low depressive symptoms tend to adaptively use reappraisal more when a situation is perceived as less controllable (thus theoretically making problem-solving more helpful for the situation). By contrast, participants with higher depressive symptoms showed the reverse correlation between perceived controllability of a situation and reappraisal use, indicating inadequate use of the ER strategy based on situational needs.

Based on these prior research and theories on ER flexibility, we first predicted that individuals with MDD may be less flexible in choosing reappraisal in response to changes in situational reappraisal affordances, relative to non-depressed individuals. In other words, the positive relationship between reappraisal affordances and choice would be attenuated in the MDD group, relative to the non-depressed controls.

Secondly, we hypothesized that depression would moderate the relationship between reappraisal choice and success, with the positive association being weaker in individuals with MDD compared to the controls. Overall motivational levels tend to be lower in individuals with depression, compared to those without depression [19]. As a result, even if individuals with depression choose to use reappraisal strategies, they may not invest sufficient effort in applying them, contrary to self-determination theory. This lack of effort may reduce the effectiveness of reappraisal strategies in alleviating negative emotions related to the situation, leading to a weaker correlation between choice and success in this group.

Lastly, the path between reappraisal affordances and success may also be modulated by depression. The emotion context insensitivity theory in depression [20] argues that individuals with depression exhibit reduced emotional responses regardless of the valence of emotional

stimuli (positive or negative), suggesting that depression is linked to inflexible emotional responses in response to changing situational contexts, as supported by empirical evidence [21]. Hence, we expected that the relationship between reappraisal affordances and success would be also attenuated in the MDD group vs. controls.

In summary, the hypotheses of the current study were as follows:

Hypothesis #1: Reappraisal affordances would predict reappraisal success through an increased choice of reappraisal.

Hypothesis #2: Depression would significantly moderate the proposed mediation model. The relationships between reappraisal affordances, choice, and success would be weaker in individuals with depression compared to those without depression (Fig. 1).

METHODS

Participants

A total of 80 university students (40 with depression and 40 without depression) participated in the current study through the psychology research participation pool at the first author's university, or a flyer on campus or online. Due to a lack of prior research on reappraisal affordances in individuals with and without depression, we were unable to perform a priori power analysis. Instead, we determined our sample size based on previous experimental research comparing reappraisal choice between these two groups [22,23], leading to our goal of recruiting 80 participants.

Participants were eligible to be included in the MDD group if they: (1) met the diagnostic criteria for current MDD based on the mood module of the Structured Clinical Interview for DSM-5 Disorders (SCID-5) [24]; (2)

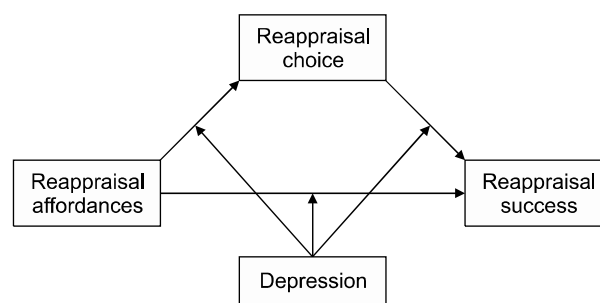


Fig. 1. The proposed moderated mediation model.

reported depressive symptoms above the clinical cut-off on the Patient Health Questionnaire-9 (PHQ-9), with a total score of 10 or above [25]; and (3) did not meet the diagnostic criteria for current or past bipolar disorder and psychotic disorders based on the Mini-International Neuropsychiatric Interview (MINI) [26]. For the non-depressed group, participants were eligible if they: (1) did not have current and past MDD, based on the SCID-5; (2) had a PHQ-9 total score below 10; and (3) did not have current or a history of bipolar disorder and psychotic disorders based on the MINI.

Clinical interviews were conducted by two clinical psychology graduate students who received training in clinical interview administration from a licensed clinical psychologist. The first three interviews conducted by each interviewer were co-rated with the licensed clinical psychologist. For these initial interviews, the student interviewer administered the interviews while the psychologist listened to the conversations via an online conference platform, with consent from the participants. In all cases, the same diagnostic conclusion was reached, and any disagreements regarding specific criteria were resolved through discussion. Among 97 students recruited, 17 were excluded due to not meeting the eligibility criteria, leading to 80 students included in the final analysis. The mean age of the participants was 20.98 with 70% being women ($n = 56$). The MDD group was significantly older than the controls, age, $t(78) = -5.92$, $p < 0.001$, and there was no significant group difference in gender, $\chi^2(1, 80) = 0.95$, $p = 0.329$. As expected, the MDD group reported greater depressive symptoms, measured by the PHQ-9, than the controls, $t(78) = 19.18$, $p < 0.001$.

Materials and Measures

Reappraisal task

To assess reappraisal affordances regarding emotional situations, we first created vignettes depicting negative events closely relevant to college students, following prior studies on reappraisal affordances that used emotional vignettes [6,9]. Because these initial studies on reappraisal affordances focused on testing whether reappraisal affordances vary by situation beyond situational intensity, they included two levels of intensity (high vs. moderate) per stimulus. Following them, 12 vignettes were initially created with six themes, each having two intensity levels.

To validate whether these emotional vignettes elicit the intended negative emotions, we conducted a pilot study in which 25 college students (mean age = 19.32, $SD = 1.30$, 17 women) were presented with each vignette and asked to rate the intensity of their negative emotional response to each vignette. Results from repeated measures analysis of variance (ANOVA) with theme and intensity as within-factor variables showed a significant main effect of intensity, $F(1,24) = 83.15$, $p = 0.004$, and the interaction effect of theme and intensity, $F(5, 120) = 3.31$, $p = 0.008$. A Bonferroni-corrected comparison analysis revealed that high intensity vignettes elicited more negative emotional response, compared to moderate intensity vignettes, as intended. Despite the significant interaction effect, the post-hoc comparison analysis did not reveal any specific vignettes that differed from others in terms of intensity (e.g., no theme elicited a significantly higher emotional response in a moderate-intensity vignette compared to other themes). To reduce participants' burden from a lengthy study procedure, eight vignettes (four themes, each with two intensities) were selected among them. Four vignettes (two themes with two intensities) were excluded because they elicited the lowest negative emotional response among high-intensity vignettes and/or the highest levels of responses among moderate-intensity vignettes. Details regarding the vignettes used in the current study and the pilot study results are presented in the Supplementary Materials A and B (available online), respectively.

The reappraisal task consisted of two parts. In the first part, participants were presented with each of the eight emotional vignettes and initially reported the first thought that came to mind regarding the given situation. They then rated the levels of their negative emotional response to the situation and perceived reappraisal affordances. To minimize the potential influence of affordance reports on their reappraisal choice, participants were asked to answer seven additional questions (e.g., how much they wanted to change their emotions in the given situation) before reporting their reappraisal choice. After completing responses to all eight vignettes, participants proceeded to the second part. They were presented with each stimulus again and instructed to use reappraisal to reduce their negative emotions by writing as many thoughts as possible for two minutes. After each two-minute reappraisal implementation, participants rated their negative emo-

tional response to the given situation again.

Negative emotional response to vignette

Participants rated the extent to which they would feel negative emotions in the given situation, if they were the person in the situation, using a 9-point Likert scale (1 = not at all, 9 = extremely).

Reappraisal affordances

Following prior research [9], participants were asked to rate how easy they thought it would be to interpret the situation differently in order to reduce negative emotions (in other words, how easy they think it would be to use reappraisal), if they were the person in the situation, on a 9-point Likert scale (1 = extremely easy, 9 = extremely difficult). This item was reverse-coded so that higher scores reflect higher levels of reappraisal affordances.

Reappraisal choice

Again, following previous research [9], reappraisal choice was assessed by asking participants to rate the extent to which they would choose reappraisal to regulate their emotions if they were the person in the situation on a 9-point Likert scale (1 = not at all, 9 = extremely).

Depressive symptoms

Depressive symptom severities were measured using the PHQ-9. Participants were asked to rate how often they have bothered by nine depression-related symptoms over the last two weeks (e.g., “feeling down, depressed, or hopeless”) on a 4-point Likert scale (0 = not at all, 3 = nearly every day). Greater total scores indicate higher levels of depressive symptoms ($\alpha = 0.93$ in the current sample).

Procedure

After providing informed consent, participants reported their demographic information, completed the PHQ-9, and were screened for eligibility via a clinical interview. Eligible participants then completed other measures unrelated to this study. They were subsequently introduced to reappraisal through written instructions that explained cognitive reappraisal and provided examples of its use. After reading the instructions, the researcher asked if participants understood what they had just read. Once participants confirmed their understanding, they completed the first and second parts of the reappraisal task in sequence.

Participants were allowed to take a short break between the two parts of the reappraisal task. This study procedure was approved by the institutional review board of Sungkyunkwan University (IRB #2024-04-031).

Data Analysis

The proposed mediation and moderated mediation models were tested using SPSS PROCESS 4.2 [27]. Throughout the analysis, the average scores of the variables across the eight vignettes were used. The mediation model in which reappraisal choice mediates the relationship between reappraisal affordances and success (hypothesis #1) was tested using Model 4. Reappraisal success, defined as decreases in negative emotional responses after reappraisal implementation, was assessed by examining negative emotional response after reappraisal implementation (post-implementation response reported in the second part of the task) as an outcome variable in the model while controlling for emotional response before implementation, reported in the first part. Thus, in the mediation model, reappraisal affordances were entered as a predictor, negative emotions post-implementation as the outcome variable, and choice as the mediator, along with negative emotions before implementation as a covariate.

A series of moderated mediation models were conducted to test depression as a moderator in the proposed mediation model (hypothesis #2). Firstly, Model 7 was used to test the moderation of depression (controls = 0, MDD = 1) in the relationship between reappraisal affordances and choice. Secondly, the moderation of depression in the reappraisal affordances and success relationship was tested using Model 5. Lastly, Model 14 was used to test moderation of depression in the reappraisal choice and success relationship. For all models, 95% confidence interval (CI) estimates were computed using the bootstrap method based on 5,000 samples. Results from correlation analyses showed that age, but not gender, was significantly associated with reappraisal affordances, $r = -0.24$, and choice, $r = -0.27$. Accordingly, age was controlled in all the models.

RESULTS

Results from independent samples *t* tests showed no significant group differences in the key variables of the study, $t(78) = 1.28$, $p = 0.992$ for reappraisal affordances,

$t(78) = 4.40, p = 0.077$ for choice, $t(78) = -0.15, p = 0.301$ for negative emotional response before reappraisal implementation, and post-implementation, $t(78) = -2.50, p = 0.405$. Mean and standard deviations of the key variables by group status are presented in Table 1, and correlations among variables are shown in Table 2.

Reappraisal Affordances Predicting Success via Choice

Results from the mediation analysis revealed that greater reappraisal affordances significantly predicted higher levels of reappraisal choice, $B = 0.30$, standard error (SE) = 0.12, $t(76) = 2.39, p = 0.019$, and reappraisal choice was significantly linked to lower levels of negative emotional response post-implementation after controlling pre-implementation response (greater reappraisal success), $B = -0.32$, SE = 0.11, $t(76) = -2.87, p = 0.005$. As predicted, based on the 95% CI, the indirect effect of reappraisal affordances on success via choice was significant, $B = -0.10$, 95% CI: $[-0.24, -0.01]$. In contrast, direct effect was not significant, $B = 0.16$, 95% CI: $[-0.09, 0.41]$, SE = 0.13, $t(76) = 1.31, p = 0.195$. That is, increased levels of reappraisal choice fully mediated the

relationship between reappraisal affordances and success.

Depression as a Moderator

Results from the first moderated mediation model revealed that depression significantly moderated the relationship between reappraisal affordances and choice, $B = 0.44$, SE = 0.22, $t(74) = 2.06, p = 0.043$. In contrast to our prediction, the conditional indirect effect was significant in the MDD group, $B = 0.51$, 95% CI $[0.20, 0.82]$, SE = 0.16, $t(74) = 3.27, p = 0.002$, but not in the controls, $B = 0.07$, 95% CI $[-0.24, 0.38]$, SE = 0.16, $t(74) = 0.43, p = 0.672$. However, these results are interpreted with caution because the 95% CI for the overall index of moderated mediation included zero, $B = -0.14$, 95% CI $[-0.39, 0.01]$. Detailed results are presented in Table 3 and Figure 2.

In addition, results from the second and third moderated mediation models showed that the moderation of depression was not significant in the choice and success relationship, $B = -0.16$, SE = 0.14, $t(73) = -0.66, p = 0.514$, and in the reappraisal affordances and success, $B = -0.42$, SE = 0.23, $t(73) = -1.86, p = 0.067$.

Results from a series of two additional analyses—(1) with gender as an additional covariate and (2) with depression symptom severity as a moderator instead of a diagnostic depression group—are presented in Supplementary Material C (available online). The main findings remained consistent across these additional analyses.

DISCUSSION

This study aimed to examine whether reappraisal affordances predict success via reappraisal choice and to further investigate whether depression moderates the relationship between these variables in the model. The results

Table 1. Demographic and clinical characteristics of the sample

	HC	MDD
Age	19.08 ± 1.33	22.88 ± 3.84
Woman (%)	30 (75)	26 (65)
PHQ-9	3.62 ± 2.31	17.32 ± 3.87
Reappraisal affordances	4.46 ± 1.06	4.14 ± 1.16
Reappraisal choice	6.33 ± 0.81	5.29 ± 1.25
Pre-implementation NE	6.36 ± 1.00	6.37 ± 0.87
Post-implementation NE	4.09 ± 1.09	4.71 ± 1.13

Values are presented as number (%) or mean ± standard deviation. MDD, individuals with major depressive disorder; HC, healthy controls; PHQ, Patient Health Questionnaire-9; NE, negative emotional response.

Table 2. Correlations among variables used in the current study

	1	2	3	4	5
1. Age	-				
2. Gender	-0.19	-			
3. Reappraisal affordances	-0.24*	0.02	-		
4. Reappraisal choice	-0.27*	0.09	0.33**	-	
5. Pre-implementation NE	0.03	0.06	-0.37**	-0.17	-
6. Post-implementation NE	0.02	0.10	-0.04	-0.30**	0.25*

Gender (0 = man, 1 = woman).

NE, negative emotional response; -, not available.

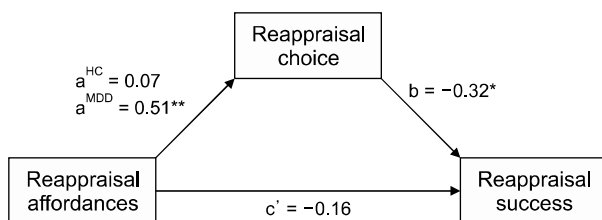
* $p < 0.05$; ** $p < 0.01$.

Table 3. Group moderating the relationship between reappraisal affordances and choice (Model 7)

Mediator model	Mediator: reappraisal choice				
	B	SE	<i>t</i>	<i>p</i>	Bootstrapping 95% CI (LL, UL)
Constant	5.59	1.21	4.63	< 0.001	(3.19, 8.00)
Affordances	0.07	0.16	0.43	0.67	(−0.24, 0.38)
Group	−1.09	0.27	−3.98	< 0.001	(−1.64, −0.54)
Group × affordances	0.44	0.22	2.06	0.04	(0.02, 0.87)
Pre-implementation NE	0.00	0.13	0.02	0.98	(−0.26, 0.27)
Age	0.04	0.04	0.88	0.38	(−0.05, 0.12)

Outcome model	Outcome variable: post-implementation NE				
	B	SE	<i>t</i>	<i>p</i>	Bootstrapping 95% CI (LL, UL)
Constant	4.37	1.47	2.98	< 0.001	(1.45, 7.23)
Affordances	0.16	0.13	1.31	0.20	(−0.09, 0.41)
Reappraisal choice	−0.32	0.11	−2.87	0.01	(−0.54, −0.10)
Pre-implementation NE	0.34	0.14	2.41	0.02	(0.06, 0.62)
Age	−0.01	0.04	−0.33	0.74	(−0.09, 0.06)

NE, negative emotional response; SE, standard error; CI, confidence interval; LL, lower limit; UL, upper limit.

**Fig. 2.** The moderated mediation model with depression as a moderator.

HC, healthy controls; MDD, individuals with major depressive disorder.

* $p < 0.05$; ** $p < 0.01$.

revealed that reappraisal choice fully mediated the relationship between reappraisal affordances and success. Depression significantly moderated the link between reappraisal affordances and choice. Unexpectedly, this relationship was only significant and positive in the MDD group, but not in the controls. Furthermore, in contrast to our hypotheses, the other paths, particularly the relationship between choice and success, were not significantly moderated by depression.

Firstly, as predicted, we found that reappraisal affordances predicted success through increased choice of reappraisal, aligning with both Tamir's [10] ER model and self-determination theory [12], which suggests that individuals are more likely to actively engage in ER strategies, such as reappraisal, when they perceive them as more feasible, leading to greater success in task perform-

ance. Our findings further revealed that reappraisal choice fully mediated the relationship between affordances and success, and the non-significant direct effect of affordances on success aligns with previous findings [28]. This suggests that perceived feasibility of using reappraisal in a given context links to success only when it predicts reappraisal choice (increased willingness to use the strategy), suggesting the potential importance of actual effort in strategy use. Future research is imperative to examine whether increased effort in strategy use is indeed the central driver of the full mediation of choice in the relationship between perceived reappraisal affordances and success.

Secondly, in contrast to our hypothesis, we found that the relationship between reappraisal affordances and choice was significant only in the MDD group, but not in the controls. While our findings underscore the importance of considering depression status in understanding the role of reappraisal affordances in choice, they are inconsistent with the ER flexibility literature, which highlights the importance of context sensitivity in emotional well-being [7,8], as well as previous research suggesting that depression is associated with less flexible choice of an ER strategy based on situational contexts [14].

One possible explanation is that reappraisal affordances may serve as a unique situational determinant of reappraisal choice among individuals with depression. Given that reappraisal affordance is a situational characteristic that is not purely situational due to its reliance on

one's perception of the cost of using the strategy in a given context, this situational factor may be more susceptible to the cognitive resources available to individuals for using the strategy. Previous research has shown that individuals with depression have reduced cognitive resources compared to their non-depressed counterparts [29,30], which can impair their ability to effectively engage in cognitively demanding strategies, such as reappraisal [31]. As a result, individuals with depression, who already have limited cognitive resources, may rely more heavily on their perception of the difficulty of using reappraisal—a cognitively taxing strategy—when deciding whether to use it. This may explain the significant relationship between reappraisal affordances and choice observed in this group in our results.

Conversely, non-depressed individuals, who generally have more cognitive resources than those with depression, may be less influenced by the perceived cognitive burden associated with ER strategies. According to Tamir [10], the motivational processes that determine whether to use an ER strategy, such as reappraisal, are shaped by an individual's evaluation of both the costs and benefits of the strategy. Non-depressed individuals may weigh both the perceived benefits and cognitive burden (cost) as well as other situational factors when choosing an ER strategy, allowing them to be more flexible in their decisions by considering multiple factors, rather than focusing predominantly on the cognitive cost. Hence, the non-significant correlation between reappraisal affordances and choice in the non-depressed group may not necessarily reflect a reduction in ER flexibility, and likewise, the significant path in the MDD group may not represent flexible ER either.

To examine whether the current results are linked to differences in cognitive resources between individuals with and without depression—particularly given the cognitively demanding nature of the reappraisal task used in this study, which involves repeated decision-making—future research should directly measure cognitive ability and/or employ tasks with varying cognitive demands. Furthermore, it is important for future studies to clarify if the unexpected finding of a significant correlation between affordances and choice only in the MDD group is specific to depression or can be generalized to other forms of psychopathology. This would help determine whether reappraisal affordances indeed serve as a unique

situational factor influencing reappraisal choice specifically in depression.

Lastly, we found no significant moderation in the other two paths, particularly in the relationship between reappraisal choice and success. On one hand, these findings suggest that individuals with depression, like those without depression, are more likely to experience reduced negative emotions in a given situation through reappraisal when they report increased willingness to use reappraisal in that context. On the other hand, participants in our study were forced to implement reappraisal in the second part of the reappraisal task. Thus, reduced negative emotional responses to situations reflect changes in emotions after forced implementation of reappraisal. According to self-determination theory, effort invested in task performance is the main reason why one's willingness to engage in a task is linked to greater success in task performance [12,19]. Because we forced effort in our reappraisal task by asking participants to implement the strategy, we cannot rule out the possibility that individuals with depression may have put more intentional effort into the implementation than they would naturally do when uninstructed.

The current study has several limitations. First, the short interval between the report of reappraisal affordances and choice per vignette may have influenced our results. To reduce its potential influence, previous studies assessed reappraisal choice one week after participants' initial reports of reappraisal affordances [6,9]. We could not implement such a strategy due to expected difficulties and concerns in recruiting and retaining clinical samples for over more than one session. Although we intentionally placed several questions between the two reports, the potential influence of reporting affordances on choice remains in our study.

The second limitation concerns the order and types of vignettes used in the reappraisal task. In the present study, vignette pairs with the same theme but differing in intensity were presented in a fixed order for all participants. Specifically, the high-intensity vignette was always presented before the moderate-intensity vignette within each theme. This fixed presentation order may have introduced order effects, such that exposure to the high-intensity vignette could have influenced participants' emotional responses to the subsequent medium-intensity vignette, potentially affecting reappraisal success. Additionally, the

vignettes used in this study were intended to elicit limited types of negative emotions, mostly sadness and anxiety relevant to college contexts (see the Supplementary Material A for details; available online). Previous research on reappraisal affordances [6,11] has used multiple types of stimuli (e.g., images, vignettes) to evoke a wider range of negative emotions. The limited range of negative emotions illustrated in our stimuli may have influenced the non-significant link between reappraisal affordances and choice in the non-depressed group.

Third, comorbid psychiatric conditions—such as anxiety and personality disorders—were not assessed or controlled for in this study. Given that ER inflexibility is a transdiagnostic feature associated with various forms of psychopathology [7,8], we cannot rule out the possibility that the observed relationship between reappraisal affordances and choice in the MDD group was influenced by comorbid symptoms rather than depression itself. Similarly, undiagnosed psychopathology in the control group may have contributed to the absence of a significant association in that group. Future research should examine whether these findings remain robust when accounting for additional clinical symptoms across both groups.

Lastly, our sample consists of college students, which may limit the generalizability of the current findings to the broader population. For example, cognitive abilities tend to decline with age [32], especially in individuals experiencing depression [33]. With reduced cognitive resources, older adults may find it more difficult to view the same situation differently (low reappraisal affordances) than younger adults do. Furthermore, their choice of strategy may be more influenced by its perceived cognitive cost, especially since the use of reappraisal was associated with temporary declines in cognitive functioning in older adults [34]. Thus, it is essential for future research using a wider sample particularly in terms of age to investigate whether the current mediation model is moderated not only by depression status but also by age and their interaction.

The current study was novel in that it extended the literature on reappraisal affordances [6] by demonstrating that one's perceived reappraisal affordances predict not only reappraisal choice but also success through increased choice. Further, our results underscore the need to consider depression in understanding this mediation, revealing that depression significantly moderated the link between reappraisal affordances and choice. Our findings

regarding the significant moderation by depression suggest that individuals with depression may be more susceptible to the perceived cost of using a cognitively taxing ER strategy, such as reappraisal. Thus, it might be important to help individuals with depression adjust their perceived burden of a situation to encourage them to use reappraisal when it is appropriate. One potential approach to reducing this perceived burden is to target cognitive function and flexibility in individuals with depression. Future research is needed to test whether cognitive remediation interventions or training aimed at enhancing cognitive functions in depression [34] can help individuals perceive emotional situations as more amenable to reinterpretation, thereby increasing their use of reappraisal. In addition, these findings raise the possibility that a stronger relationship between situational factors and ER strategy choice, which is often used as an indicator of context-sensitivity [35], may not always reflect ER flexibility.

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■ Conflicts of Interest

No potential conflict of interest relevant to this article was reported.

■ Author Contributions

Conceptualization: GyuSeok Choi. Formal analysis: GyuSeok Choi. Funding acquisition: Sunkyoung Yoon. Methodology: GyuSeok Choi. Project administration: Eunseo Soh. Supervision: Sunkyoung Yoon. Writing—original draft: GyuSeok Choi. Writing—review & editing: Eunseo Soh, Sunkyoung Yoon.

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